Muhammad Tanseef Shahid

2D/3D Computer Vision & AI Engineer

mtanseefshahid@gmail.com tanseefshahid.github.io linkedin.com/in/tanseef
github.com/tanseefshahid

Computer vision engineer and AI researcher with **6+ years** of production experience building end-to-end 2D/3D vision systems — from industrial 3D laser inspection and point cloud metrology to generative AI, real-time edge detection, and autonomous hardware integration. Published researcher (IEEE Access, CMMM) with M.S. in Computer Vision. Proven track record delivering sub-200ms real-time inference, 98%+ accuracy detection systems, and high-precision sub-millimeter 3D measurement pipelines deployed on factory floors.

Core Competencies

Computer Vision & 3D

- Object Detection & Tracking (YOLO, DeepSORT, Mask R-CNN)
- Image Segmentation (U-Net, U2Net, SAM, Active Contours)
- 3D Reconstruction (NeRF, OpenLRM, Point Cloud to Mesh)
- 6-DoF Pose Estimation (FoundationPose, ICP, Hand-Eye Calib.)
- 3D Metrology (RANSAC, PCA, FAI, Coordinate Alignment)
- Generative Models (Latent Diffusion, GANs, Textual Inversion)

Software & Tools

- Python, C++, C# (.NET 8), TypeScript, MATLAB
- PyTorch, TensorFlow, OpenCV, PCL, Open3D
- Blender (bpy), PyVista/VTK, Eigen, Helix Toolkit
- Docker, Git, ONNX, CUDA, Linux
- MFC, WPF, PyQt, Flask, React/Next.js
- NVIDIA Jetson, Basler GigE, PLC/GPIO, AWS

Professional Experience

3D Metrology & AI Engineer — Aug 2025 – Present
Hyvision System — Seoul, South Korea

- Architected a high-speed 3D laser inspection system for UMP modules using **PCL** and **RANSAC/Eigen**-based coordinate alignment, achieving sub-millimeter measurement precision with automated FAI validation against SQL Server specifications.
- Developed a PolyWorks-style 3D alignment application with **ICP best-fit**, landmark registration (SVD), and interactive lasso-based noise removal — processing point clouds of 1M+ points in real-time using **VTK/PyVista**.
- Designed a modular 3D rendering engine with a visual **node-based workflow editor** (ECS architecture, .NET 8/WPF), enabling non-programmers to build custom inspection pipelines via drag-and-drop with an extensible plugin system.

Computer Vision Engineer — Apr 2025 – Jul 2025
Luxolis — Seoul, South Korea

- Deployed a real-time 360° defect detection system using a synchronized **6-camera Basler GigE array** with YOLO inference, achieving **<200ms end-to-end latency at 5 parts/sec**. Integrated with PLCs via NuDAQ I/O for pneumatic rejection of defective units.
- Built an **eye-in-hand 6-DoF pose estimation** pipeline for autonomous CNC machining using an RGB-D camera, FoundationPose (zero-shot), and ICP refinement — eliminating manual fixturing and enabling dynamic toolpath adjustment in SE(3) space.

2D/3D AI Researcher — Apr 2022 – Apr 2025
PERSPECTIVE Corp. — Seoul, South Korea

- Architected a photorealistic **virtual try-on** pipeline using LADi-VTON (Latent Diffusion), DensePose, U2Net cloth segmentation, and CLIP textual inversion, achieving **85%+ pattern fidelity**. Deployed on AWS with Docker.
- Created a single-image to **3D mesh reconstruction** pipeline using OpenLRM, achieving IoU 0.80 and Chamfer Distance 0.08 with 24×360° Blender-rendered training views enhanced by ESRGAN.
- Designed a hierarchical **apparel classification** system (ResNet50) achieving **>95% accuracy across 74 classes** using 3-depth taxonomy with specialized sub-classifiers.
- Led face processing R&D: face reconstruction, audio-driven lip synthesis, and face enhancement using GANs and diffusion models.

AI Developer
Ellexi

Nov 2019 – Oct 2021
Seoul, South Korea

- Developed a real-time **ALPR system** for gas station automation using YOLO + CRNN with synthetic license plate data, achieving **98% recognition accuracy** with DeepSORT tracking.
- Deployed a **PPE compliance detection** system on **NVIDIA Jetson TX2** for construction sites with multi-class YOLO inference and real-time geo-fenced hazard zone alerts.
- Built an **anomaly detection & multi-camera tracking** system for Incheon Airport using YOWO (3D-CNN), HRNet pose estimation, and Person Re-ID — detecting falls, sprinting, and abandoned baggage.
- Created a **gas meter inspection** pipeline using CRAFT + Tesseract OCR with Mask R-CNN segmentation, digitizing date stamps from weathered boiler plates.

Education

M.S. in Computer Science (Computer Vision)
Chung-Ang University (CAU)

Mar 2017 – Sep 2019
Seoul, South Korea

- Thesis: Object Segmentation with Active Contours Driven by Weight Matrix
- Research Focus: Active Contours, Image Segmentation, Object Detection & Tracking
- CAYSS Scholarship recipient (Chung-Ang Young Scientist Scholarship)

B.S. in Electronic Engineering
GIK Institute

2011 – 2016
Topi, Pakistan

- FYP: Smart Blind Navigation System (Raspberry Pi, OpenCV, GSM)
- Secured 2nd position in the final year project competition